AREEHA ZULFIQAR CHAUDHARY

Applied AI Engineer | LLM Systems - RAG - Agents - Evaluation | Python - TypeScript
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SUMMARY

Full-stack engineer specializing in LLM applications, RAG, agent workflows, and the evaluation systems that make them shippable. Built AI tooling that turns prompts and requirements into validated code changes, plus production web apps with billing, payments, and ERP integrations.

TECHNICAL SKILLS

Languages: Python | TypeScript | JavaScript | SQL | C++
AI Engineering: LLM APIs (OpenAI, Anthropic) | RAG | LangGraph agents | MCP | Embeddings & vector search (Chroma) | LLM evals & eval harnesses | LLM-as-judge | Model benchmarking | Structured outputs & tool use | Prompt engineering | Computer vision (MediaPipe, OpenCV) | scikit-learn | pandas
Frontend: Next.js | React | TypeScript | TanStack React Query | Tailwind CSS | Vite
Backend: Node.js | Express | FastAPI | REST APIs | WebRTC | Socket.IO | GraphQL | Auth
Data & Cloud: PostgreSQL | Supabase | Prisma ORM | Stripe | Odoo | Render | Railway | Vercel | AWS Amplify
Engineering Practices: Git | CI/CD (GitHub Actions) | Docker | pytest | Jest | Playwright | Observability | Agile / Scrum | Code review | Technical documentation | Claude Code

EXPERIENCE

Junior Software Engineer - **NewGates Systems** *| Lahore, Pakistan | Remote delivery for US clients*
Feb 2026 - Present

AI Developer Tooling

- **Built Prototype Studio, a requirements-to-prototype workspace**: client prompts become tracked requirements, attached specs and acceptance checks guide AI edits, and failed changes never replace the working version.
- **Designed the AI code-edit safety pipeline**: forced structured file changes, server-side path whitelisting, snapshot and staging copies, esbuild validation, atomic promote, and one-click restore.
- **Built self-correcting design and code generation loops**: Playwright renders outputs, pixelmatch and vision judging score Figma matches, lower-cost models draft React changes, Claude reviews against production, approved work opens PRs with rollback and cost tracking.

Member Portal - US Community Solar Cooperative

- **Drive day-to-day feature delivery on a live production portal**: build, test, document, and demo client-facing changes directly to a US client across time zones.
- **Delivered billing and payment workflows**: Stripe card and ACH tokenization, saved payment methods, automated invoice payment, invoice PDFs, multi-step onboarding, and Odoo ERP integration.

PROJECTS

Roamio - AI Trip Planner *Next.js | FastAPI | LangGraph | Chroma | OpenAI | Tavily | 2026*

- **Built a RAG and agent trip planner as a LangGraph state machine** with search, planning, re-planning, live conditions and writing stages, turning a few constraints into a grounded day-by-day itinerary.
- **Designed it so deterministic Python owns every trusted decision** including route, cost, and feasibility, while the LLM only writes prose.
- **Grounded every generated note in two retrieval pipelines** behind an LLM-as-judge faithfulness guard, then measured it: 91% judge-verified, 14s median and ~$0.001 per itinerary across a benchmark suite.
- **Took it to production and exposed it over MCP**: Docker container deployed on Render, 41 pytest tests and the image build gated by GitHub Actions, request-traced logging, and the planning tools published as typed MCP tools.

SmartMeet - AI Meeting Analytics (Final Year Project) *Next.js | Node.js | GraphQL | WebRTC | Socket.IO | MediaPipe | 2026*

- **Engineered a real-time meeting analytics system** that extends video meetings with face-presence, gaze, and head-pose analysis to estimate participant engagement. Led frontend delivery and the MediaPipe CV integration.
- **Also built - Uraan**, an AI business-roadmap platform for women entrepreneurs (Next.js, Prisma/PostgreSQL, NextAuth, OpenAI), and **VoltKar**, a smart energy tracker with a Jest test suite (Next.js, TypeScript, Prisma). Both live.

CERTIFICATIONS

Hugging Face — AI Agents Course**,** MCP Course *Sept–Oct 2026*
Kaggle Learn — Intermediate Machine Learning**,** Intro to Deep Learning**,** Pandas *Oct 2026*

EDUCATION

BS Software Engineering - Punjab University College of Information Technology (PUCIT), Lahore *| 2022 - 2026*

- **Teaching Assistant, Artificial Intelligence** - supported 30+ students in AI concepts, algorithms, and assignments under faculty supervision.